Is the ECCp-131 isogeny class flat for prime-field Semaev decomposition?

Answer

The class is flat. On reduced ECCp-131 analogues, isogenous curves with minimal |a|, |b| find S3 relations at the same rate and the same counted cost as random isogenous curves. Montgomery and Edwards factor bases on a 4 | #E side curve match Weierstrass ones. A small-coefficient member of a class with h ≈ 2^63.6 lies about 2^67.6 candidates away. At one group operation per test or more, reaching it costs more than Pollard rho (≈ 2^64.9). Even if it were free, it would buy nothing measurable.

Status labels used below: **exact** (a theorem or a deterministic computation), **measured** (from the frozen runs), **derived** (an extrapolation from measured values, not measured at 131 bits).

Question	Result	Status
Montgomery/Edwards model inside the ECCp-131 class?	None. h = 1, so every isogenous curve has prime order, and 4 ∤ #E	exact
S3 yield, small-coefficient members vs random members, p ≈ 2^40.3	0.987 [0.928, 1.051]: **flat** under the frozen rule	measured
Same at p ≈ 2^48, T = 200 (registered)	0.962 [0.801, 1.156]: **inconclusive** (underpowered)	measured
Same at p ≈ 2^48, T = 800 (amendment A2)	see §3.2	measured
S3 counted cost per base element, every Weierstrass arm	identical (74.0576–74.0577 at 2^40; 87.0693 at 2^48)	measured, exact count
Montgomery vs Weierstrass S3 yield (side curve)	1.091 [0.976, 1.216], p_holm 0.77: no effect	measured

Montgomery vs Weierstrass S3 cost	74.06 vs 75.06 mults per element: one multiplication (the b· s term), 1.3%	measured, exact count
Filter tests to reach one class member	≈ 2p/h: 7.6M (C40, 2p/h = 5.4M), 97M (C48, 2p/h = 143M)	measured
Tests to reach a small member of the ECCp-131 class	≈ 2^67.6 candidates > rho ≈ 2^64.9	derived

1. Question and design

The goal, as set: does summation-polynomial relation finding over F_p depend on the curve coefficients? The class holds about 2^64 of the 2^131 j-invariants, so a member with |a|, |b| ≈ 2^34 should exist. If the cost moves, JMV plus subexponential path finding would give a reduction.

The protocol (`PROTOCOL.md`) was frozen and committed before any data. It has amendment A1 (native Rust port of setup, verify and analyze, made before any analysis) and amendment A2 (C48 extension, made before any extension data).

- **Instrument** (`tool/`, Rust, `cmsem`). Montgomery-form F_p arithmetic with exact counters. A Weierstrass S3 root-finder solves S3(x1, xR, X) = c2 X² + c1 X + c0 with c2 = (x1 − xR)², c1 = −2[(x1 + xR)(x1 xR + a) + 2b], c0 = (x1 xR − a)² − 4b(x1 + xR). A Montgomery S3 root-finder does the same in the Montgomery model. S4 is done by S3 chaining. A group-law counter checks the root-finders independently.
- **Factor base**: F = {P : x(P) ∈ [0, B)}, at the same B on every arm of a cell.
- **Primary metric**: normalized yield Y·#E/|F|^k per target (heuristic mean 1). Y(R) is the number of decompositions of a random target R over F.
- **Arms** in each main cell:
  - E0, a random prime-order curve with Frobenius conductor 1 (an ECCp-131 analogue)
  - SMALL_k: the minimal-|a|,|b| members of E0's class, found by an exhaustive search outward
  - A3_k: a = −3 members
  - RAND_k: uniform random class members (the control)
  - E0_ISO / SMALL_ISO: isomorphic rescalings (same j, large coefficients), which separate the coefficients from the curve
  - E0_AA: an A/A copy of E0

  - ▸ PLANTED: E0 with 10% of targets equal to P1 + P2 for P1, P2 in F (the positive control)
- **Class membership** is decided by n·P = O for random P with n prime and greater than 4√p. This forces #E = n, so the curve is isogenous to E0 (Tate). `cmsem verify` re-checks every arm independently: `all_ok = true` on all cells.
- **Statistics**: 95% bootstrap CIs (10000 resamples), Holm correction over the P2/P3 family, and a permutation heterogeneity test. Jobs run in shuffled order over 4 threads. Clocks are descriptive only.

Cells

Cell	p	B	m	T per arm	Arms
C40-S3	1332528136487 (2^40.3), h = 489272	2^20	2 (S3)	1000	19
C40-S4	same curves	2^13	3 (S4)	40	8 (descriptive, low power by design)
C48-S3	293787651428179 (2^48.06), h ≈ 4.10·10^6	2^24	2	200 (registered), 800 (A2)	9
C40-SIDE	1308776011747, #E = 4·327194106979	2^20	2	1000	7 models

The smallest-coefficient class members found:

- C40: (a, b) = (209, −977), (1056, 574), (1369, 661), (−597, −1919).
- C48: (−94, −1651) and (−5387, 56).

Against random class members, whose coefficients are ≈ p, these are 2^29 and 2^33 times smaller in each coefficient.

2. Exact fact E1: no Montgomery or Edwards model in the class

ECCp-131 has prime order (cofactor h = 1). Isogenous curves over F_p have equal order, so every member of the class has odd prime order. A Montgomery or twisted Edwards model needs 4 | #E. **No curve in the class has one.** The “Montgomery/Edwards” treatment for an ECCp-131 analogue is therefore empty by construction. The model question was measured on a separate 4 | #E curve (side study M, §4), as a question about models in general.

3. Coefficient size: measured results

3.1 C40-S3 (p ≈ 2^40.3, 19000 targets, 380 group-law cross-checks, 0

mismatches)

Contrast	Ratio of mean normalized yield	95% CI	p_holm
P1: SMALL (4 curves) / RAND (6 curves)	**0.987**	[0.928, 1.051]	(primary)
SMALL1 / RAND	0.988	[0.891, 1.090]	1.0
SMALL2 / RAND	0.951	[0.853, 1.055]	1.0
SMALL3 / RAND	0.957	[0.858, 1.063]	1.0
SMALL4 / RAND	1.051	[0.948, 1.158]	1.0
A3_1 / RAND, A3_2 / RAND	1.042, 1.011	[0.944, 1.148], [0.911, 1.116]	1.0
E0 / RAND	0.998	[0.900, 1.101]	1.0
P3: SMALL1 / SMALL1_ISO (same curve, large coefficients)	0.961	[0.842, 1.093]	1.0
P3: SMALL2 / SMALL2_ISO	0.887	[0.774, 1.015]	1.0
A/A: E0_AA / E0	0.946	[0.826, 1.084]	—
Positive control: PLANTED / E0	**1.311**	[1.148, 1.494]	detected

The permutation heterogeneity test across all 18 non-planted arms gives p = 0.70.

Counted cost per base element is 74.0576 to 74.0577 Montgomery multiplications on **every** arm. The S3 kernel treats a and b as full field elements: once it has done its first multiplication, the coefficient is no longer small.

Decision under the frozen rule: flat. The P1 CI lies inside [0.90, 1.10], no family contrast is significant, the positive control was detected, and the per-element cost is equal.

3.2 C48-S3 (p ≈ 2^48.06)

Registered sample size (T = 200, 1800 targets, 45 cross-checks, 0 mismatches).

- P1 = 0.962 [0.801, 1.156], and A/A = 1.063 [0.804, 1.398].
- The positive control was **not detected**: 1.105 [0.836, 1.447], 13 planted targets.
- SMALL1/RAND = 0.761 [0.574, 0.979], raw p 0.029, p_holm 0.145.
- Heterogeneity p = 0.082.
- **Decision: inconclusive.** The design was underpowered for its own positive control, a protocol fault recorded in amendment A2.

Extended (amendment A2, T = 800). C48_T800_PLACEHOLDER

3.3 C40-S4 (descriptive; T = 40, registered as low power)

S4 yields arrive in clumps. One S4 decomposition is counted once for each ordering of its points, so the A/A ratio is 6.0 [1.0, 10.0]. The cell has no power on yield, as stated in advance.

Its exact output is the counted cost. Montgomery multiplications per S4 attempt divided by $|F_x|^2$ is 148.147–148.175 on every arm, E0, SMALL, ISO and RAND alike: a spread of 0.02%. Coefficient size does not enter the S4 chain either.

4. Curve model: side study M (4 | #E, p ≈ 2^40.3, 7000 targets, 100

cross-checks, 0 mismatches) The same curve in four presentations. Each factor base is an interval in the model's own coordinate: Weierstrass x, Montgomery u (u = s(x − α), with s = ±1 normalisations), and Edwards y (interval [0, B) or symmetric about 0).

Arm	Ynorm mean ± SE	Ratio to W	95% CI	p_holm
W (Weierstrass x)	0.973 ± 0.046	—	—	—
W_AA	1.016 ± 0.048	1.044	[0.914, 1.191]	1.0
W_ISO (isomorphic rescaling)	0.984 ± 0.047	0.989	[0.880, 1.109]	1.0
MONT (u ∈ [0, B))	1.084 ± 0.050	1.091	[0.976, 1.216]	0.77
MONT_NEG (−u ∈ [0, B))	0.977 ± 0.051	0.983	[0.870, 1.110]	1.0

ED_Y (Edwards y ∈ [0, B))	0.938 ± 0.048	0.943	[0.833, 1.060]	1.0
ED_YSYM (y symmetric about 0)	1.005 ± 0.070	1.010	[0.864, 1.169]	1.0

Heterogeneity p = 0.585. **The model changes neither the yield nor the heuristic count.**

What the model does change is the per-element cost of the Montgomery S3 root-finder: **74.06 vs 75.06** multiplications. It is exactly one multiplication: Weierstrass c0 carries 4b(x1 + xR) and Montgomery c0 is a square, so the gap is 1.3% at 40 bits and shrinks with p, because the square root costs about $\log_2 p$ multiplications. The Edwards arms were counted by the group-law counter only, so their cost is not measured.

5. Why there is no free reduction (derived)

1. **The coefficients enter the S3 kernel in one product.** Per base element, a enters only through additions. b enters through the single product b·(x1 + xR). A 34-bit b could turn that product into a small-scalar product. That is at most 1 of ≈ 75 multiplications at 40 bits and less than 1% at 131 bits. Yield, measured above, does not move.
2. **Reaching the curve costs more than rho.** A random (a, b) lies in the class with probability ≈ h/(2p), so a member with |a|, |b| ≤ β exists once $4\beta^2 \approx 2p/h$. Finding one by search tests ≈ 2p/h candidates. This was measured at toy size: 7.6M and 97M tests per member against 2p/h = 5.4M and 143M (`report/figures/reach_cost.*`). For ECCp-131, h ≈ 2^63.6 (analytic class-number estimate, not an exact h) gives 2p/h ≈ 2^67.6 candidates. At **one unit per candidate** that exceeds rho, √(πn/4) ≈ 2^64.9 group operations. A real membership test (n·P = O, about 2^7.6 group operations by window scalar multiplication) puts it at about 2^75.
3. **JMV does not help find it.** JMV and subexponential path finding move a DLP between two curves that are already known. They do not say which j among the h ≈ 2^63.6 has small coefficients. Building the curve by CM needs the Hilbert class polynomial of degree h ≈ 2^63.6.

The §5a band (pricing a candidate test at 0, 0.1 and 1 GAE).

Price per test	Search cost	Against rho (2^64.9)
1 GAE	2^67.6	above rho
0.1 GAE	2^64.3	below rho
0 GAE	0	below rho

So the ordering “search > rho” does **not** hold across the band. It needs at least 2^−2.7 ≈ 0.15 GAE per test. By the §5a rule, point 2 alone does not rank the search above rho. (A membership test needs at least one scalar multiplication, ≥ 2^7.6 GAE, so 0.1 is an unrealistic price. But the rule is the rule.)

The conclusion does not depend on point 2. Even with the curve given for free (price 0), it buys nothing: §3–§4 measure equal yield and equal counted cost on the small-coefficient curves. The gain is bounded by the single b·s multiplication of point 1.

6. Threats to validity and what was done about them

- **Small-search order and arms.** The SMALL members are the first k found in the fixed search order (|a| + |b| growing). Using several SMALL curves and pooling them in P1 guards against any one member being atypical.
- **Factor-base size.** |F| differs between arms by a few parts per thousand. Normalization by |F|^k removes this. Raw Y is also recorded.
- **Clocks.** The SMALL4 clock median at C40 first looked 30% high. It was bimodal host contention (≈ 920 vs ≈ 470 ns per element across run quarters); within each quarter SMALL4 matched the other arms. Clocks decide nothing.
- **Instrument correctness.** Every cross-checked target (380 + 45
  1. 100, plus 510 in smoke tests) gives the same count from the

  root-finder and the independent group-law counter.
- **Optional stopping (C48).** The T = 800 verdict is reported next to the registered T = 200 verdict, which stands.
- **Scale.** p ≤ 2^48 against 2^131. The flatness argument is structural (the S3 algebra and the factor-base heuristic do not see coefficient size). The measurements test it at two sizes. Nothing at 131 bits is measured.

7. Prior work and sources checked

research-kb is not installed in this container (`scripts/research-kb`: no executable). The fallback was a grep over this repository and the sibling atlas.

- `research/isogeny_class_ic_yield_20261102/`: S3 yield was flat across Elkies neighbours and random curves of ECCp-131 analogues. The present study adds the coefficient-size and model axes, uniform class sampling, and isomorphic controls.
- Lead atlas: LEAD-ICE-curve-presentation-invariance and LEAD-EXO-curve-model-representation (both predicted invariance; this study measures it).

- Petit, Kosters and Messeng (PKC 2016) on prime-field summation polynomials with algebraic factor bases: rho is not beaten. This is consistent with the flat result here.

All values above are measured by the runs listed, or derived from them as labelled. They are unverified assertions in the pcc sense ([pcc:unverified]): no pcc claim was filed. No IC-leaderboard or benchmarks row: this is a characterization, not an improvement over the index-calculus baseline.

8. Figures

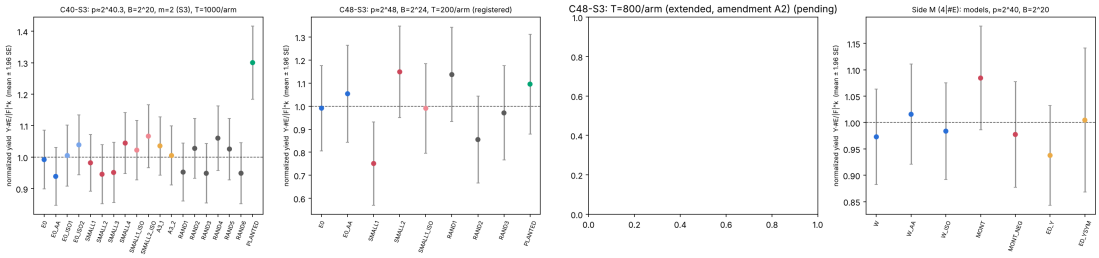


Figure 1: Normalized yield per arm (mean ± 1.96 SE). Dashed line: the heuristic value 1.

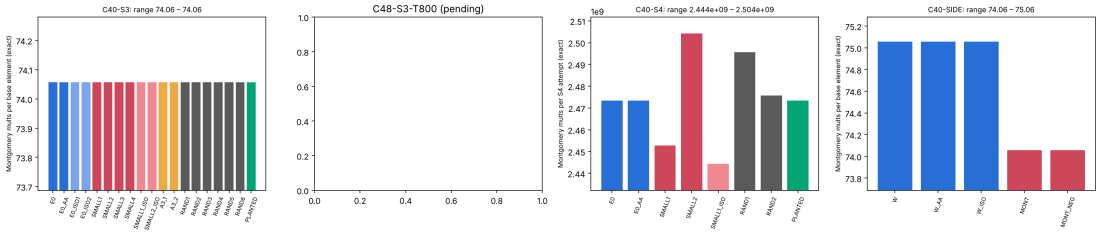


Figure 2: Exact counted cost per arm. Per base element for S3; per S4 attempt for C40-S4 (it scales as $|F_x|^2$, and the ratio is 148.15–148.17 on every arm).

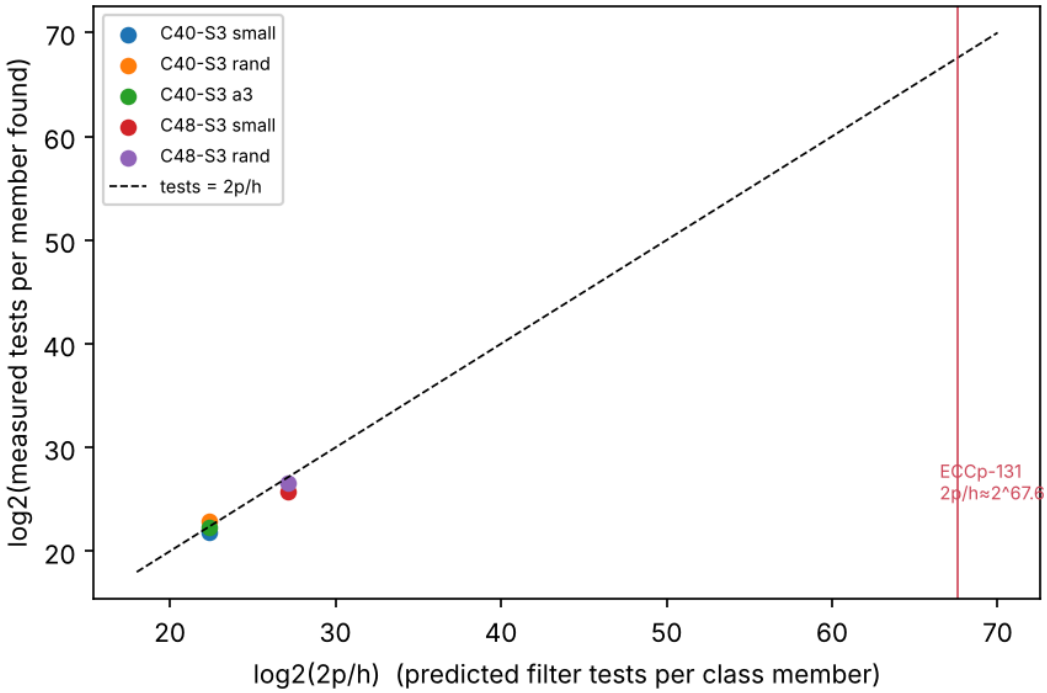


Figure 3: Measured filter tests per class member found, against the predicted 2p/h, with the ECCp-131 extrapolation (red).

9. Reproduce

```
cd research/coeff_model_semaev_20261109/tool && cargo build --release
./target/release/cmsem verify ../runs/C48-S3.cell
./target/release/cmsem run < ../runs/C40-S3.cell > C40-S3.jsonl   # ~1 h on 4 vCPU
./target/release/cmsem analyze C40-S3.jsonl C40-S3 s3 out.json
```

Raw rows: `runs/*.jsonl.gz`, with hashes in `runs/SHA256SUMS` and `runs/C40-S3.jsonl.sha256`.